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The Effects of Lead and Copper on the Cellular Architecture and Metabolism of the Red Alga *Gracilaria domingensis*

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Abstract: The effect of lead and copper on apical segments of *Gracilaria domingensis* was examined. Over a period of 7 days, the segments were cultivated with concentrations of 5 and 10 ppm under laboratory conditions. The samples were processed for light, confocal, and electron microscopy, as well as histochemistry, to evaluate growth rates, mitochondrial activity, protein levels, chlorophyll *a*, phycobiliproteins, and carotenoids. After 7 days of exposure to lead and copper, growth rates were slower than control, and biomass loss was observed on copper-treated plants. Ultrastructural damage was primarily observed in the internal organization of chloroplasts and cell wall thickness. X-ray microanalysis detected lead in the cell wall, while copper was detected in both the cytoplasm and cell wall. Moreover, lead and copper exposure led to photodamage of photosynthetic pigments and, consequently, changes in photosynthesis. However, protein content and glutathione reductase activity decreased only in the copper treatments. In both treatments, decreased mitochondrial NADH dehydrogenase activity was observed. Taken together, the present study demonstrates that (1) heavy metals such as lead and copper negatively affect various morphological, physiological, and biochemical processes in *G. domingensis* and (2) copper is more toxic than lead in *G. domingensis*.

Key words: *Gracilaria domingensis*, heavy metal, morphology, photosynthetic pigments, biochemical response

Introduction

Over the past few years, increasing human population and industrial development have led to an increase of contaminants in aquatic systems (Rocchetta et al., 2007). Accordingly, studies reporting the effects of heavy metals on aquatic organisms are currently attracting more attention, particularly those focusing on industrial and urban pollution. The contamination of coastal waters with trace metals through sewage and other anthropogenic sources has become a severe problem (Mamboya et al., 1999). Heavy metals, such as lead, copper, cadmium, zinc, and nickel, are among the most common pollutants found in both industrial and urban effluents (Sheng et al., 2004). In low concentrations, some heavy metals (Cu, Zn, Ni, and Mn) are essential trace elements for photosynthetic organisms; however, in higher concentrations, these metals cause severe toxic effects (Hu et al., 1996). Heavy metals affect all biological organisms, especially those in the aquatic ecosystem. Several studies have shown such effects as decreased macroalgal growth rates (Mamboya et al., 1999), photosynthetic pigments (Rocchetta et al., 2007; Bouzon et al., 2012), and photosynthetic efficiency (Mamboya et al., 1999; Bouzon et al., 2012) but increased total proteins and lipid contents (Rocchetta et al., 2007). Some reports have shown changes in the chloroplast ultrastructure of the red algae *Audouinella savina* (F.S. Collins) Woelkerling (Talarico, 2002), *Ceramium ciliatum* (J. Ellis) Ducluzeau (Diannelidis & Delivopoulos, 1997), *Gracilaria domingensis* (Kützing) Sonder ex Dickie (Santos et al., 2012), and *Hypnea musciformis*

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(Wulfen) Lamouroux (Bouzon et al., 2012); the green algae *Dunaliella minuta* Lerche (Visviki & Rachlin, 1992) and *Enteromorpha flexuosa* (Wulfen) J.Agardh (Andrade et al., 2004); the photosynthetic euglenoid *Euglena gracilis* Klebs (Rocchetta et al., 2007); and the brown alga *Padina gymnospora* (Kützing) Sonder (Andrade et al., 2002).

The gradual increase in the discharge of heavy metals and other pollutants into the environment directly exposes marine organisms to different levels of toxicity, affecting development and decreasing both growth and biodiversity (Torres et al., 2008). Heavy metals in high concentrations are nonbiodegradable pollutants (Mallick & Rai, 2001).

Lead (Pb) is a major environmental contaminant that arises from human, agricultural, and industrial activities, e.g., mining, burning of coal, effluents from storage battery manufacture, automobile exhaust, metal plating and finishing operations, fertilizers, pesticides, and additives in pigments and gasoline (Eick et al., 1999). This metal is not an essential element for photosynthesis in an organism, but it can be easily absorbed and accumulated in different parts of the organism (Sharma & Dubey, 2005).

On the other hand, copper (Cu) is an essential micronutrient for plant growth and development. This metal is a structural element in regulatory proteins and participates in photosynthetic electron transport, mitochondrial respiration, oxidative stress, cell wall metabolism, transcription, protein trafficking, and hormone signaling (Yruela, 2005). However, in excess, it can inhibit growth and impair important cellular processes, such as photosynthetic electron transport, photosynthesis, and respiration. The membrane transport systems seem to be Cu targets, playing a central role in toxicity processes (Yruela, 2005). Various sources of copper, including industrial and domestic wastes, agricultural practices, copper marine drainage, copper-based pesticides, and antifouling paints, have led to a clear increase in Cu concentrations in aquatic environments (Callow & Callow, 2002).

Gracilaria is a red macroalga (Rhodophyta, Gracilariaceae) extensively used in industry based on its production of secondary metabolites (Lee et al., 1999), and the genus is responsible for about 60% of the world's agar production (Tseng, 2001). The agar producer species *G. domingensis* is important in the human diet (fresh food), and Japan occupies a prominent place in the consumer world market (Guimarães et al., 1999). This species is distributed along the Brazilian coastline from Ceara State to Santa Catarina State (Schmidt et al., 2010*a*). It occurs within the upper intertidal zone to the subtidal zone.

Despite the ecological and economic importance of the red macroalga *G. domingensis*, the effects of copper and lead exposure on its biochemical and ultrastructural traits have not been studied. Thus, in this study, we investigated the *in vitro* effect of heavy metal on this species. To accomplish this, we investigated (1) changes in cellular architecture, ultrastructure, and cytochemistry between the control samples and lead and copper-treated in relation to the sensitivity of *G. domingensis* to heavy metals and (2) differences in the content of photosynthetic pigments, protein levels, antioxidant activity, and mitochondrial function after exposure to lead and copper.

Materials and Methods

Algal Material

Individuals of *G. domingensis* were collected at Ponta das Canas Beach (27°23′34″S and 48°26′11″W), Florianopolis-SC, Brazil, in July 2011, from the intertidal rocky shore and immediately transported at ambient temperature in dark containers to the Macroalgae Laboratory, Federal University of Santa Catarina, Florianopolis, Santa Catarina, Brazil (LAMAR-UFSC).

Macroepiphytes from macroalgal samples were meticulously eliminated by cleaning with a brush and filtered seawater. Apical portions were maintained in culture medium with filtered seawater plus von Stosch enrichment solution at half strength (VSES/2; Edwards, 1970) and cultivated under laboratory-controlled conditions during 14 days (experimental acclimation period) before experimental treatment with lead and copper.

Culture Conditions

The apical thallus portions were selected (±2.0 g) from the acclimated *G. domingensis* plants and cultivated for 7 days in Erlenmeyer flasks containing 500 mL of natural sterilized seawater, 34 practical salinity units (p.s.u.), and enriched with VSES/2 (without ethylene diamine tetraacetic acid). Culture room conditions were 24°C, continuous aeration, 80 μmol photons $m^{-2} \cdot s^{-1}$ [fluorescent lamps, Philips C-5 Super 84 16W/840, Brazil; LI-COR light meter 250 (LI-COR Bioscience, Lincoln, NE, USA)], and 12 h photocycle (starting at 8 h).

Experimental treatments were carried out with a control and copper and lead treatments, adding $CuCl_2$ or $PbCl_2$ (individually) at different concentrations of 5 or 10 ppm to the culture medium following the description of Tonon et al. (2011) for *Gracilaria tenuistipitata*, *Gracilaria birdiae*, and *G. domingensis*. Eight replicates were made for each experimental group.

Growth Rates

Growth rates (GRs) for treatment groups and control were calculated using the following equation: GRs [% day^{-1}] = $[(W_t/W_i) - 1] * 100/t$, where W_i = initial wet mass, W_t = wet mass after 7 days, and t = internal time in days (Penniman et al., 1986).

Light Microscope

Samples, approximately 5 mm in length, were fixed in 2.5% paraformaldehyde in 0.1 M (pH 7.2) phosphate buffer overnight (Schmidt et al., 2009). The samples were subsequently dehydrated in increasing series of ethanol aqueous solutions. After dehydration, the samples were infiltrated with Historesin (Leica Historesin, Heidelberg, Germany). Microtome sections (Leica RM 2135, Heidelberg, Germany) of

5 μm in length were stained with different cytochemical techniques. Five samples of each treatment were sectioned, and 10 cross sections per glass slide were utilized for cytochemical staining (six glass slides per cytochemical test). To evaluate mucilage layer and cell wall thickness (μm), 10 cortical and subcortical cells were measured on six different glass slides ($n = 60$) stained with TB-O. The samples were investigated with an Epifluorescent (Olympus BX 41, Tokyo, Japan) microscope equipped with Image Q Capture Pro 5.1 software (QImaging Corporation, Austin, TX, USA).

Cytochemical Staining

Light microscope (LM) sections were stained with Periodic Acid-Schiff (PAS) to identify neutral polysaccharides (Schmidt et al., 2010*a*) and TB-O 0.5%, pH 3.0 (Merck, Darmstadt, Germany) to identify acid polysaccharides through a metachromatic reaction (Schmidt et al., 2010*a*). Controls consisted of applying solutions to sections without the staining component (e.g., omission of periodic acid application in the PAS reaction).

Confocal Laser Scanning Microscopy

Live cells of *G. domingensis* samples were investigated by a laser scanning confocal microscope (Leica TCS SP-5, Wetzlar, Germany). A Leica HCX PLAPO lambda 63×/1.4–0.6 oil immersion objective was fitted on the inverted fluorescent microscope (Schmidt et al., 2012*b*). Chloroplast autofluorescence was observed at 488 nm laser wavelength excitation with emission spectrum from 639 to 701 nm (Hepler & Gunning, 1998). The LAS-AF Lite software (Leica) was used for processing the confocal images.

Transmission Electron Microscope

For observation under the transmission electron microscope (TEM), samples approximately 5 mm in length were fixed with 2.5% glutaraldehyde in 0.1 M sodium cacodylate buffer (pH 7.2) plus 0.2 M sucrose overnight. The material was post-fixed with 1% osmium tetroxide for 4 h, dehydrated in a graded acetone series, and embedded in Spurr's resin. Thin sections were stained with aqueous uranyl acetate followed by lead citrate. Four replicates were made for each experimental group; two samples per replication were then examined under TEM JEM 1011 (JEOL Ltd., Tokyo, Japan, at 80 kV). Similarities based on the comparison of individual treatments with replicates (five grids per treatment) suggested that the ultrastructural analyses were reliable.

Scanning Electron Microscope

The samples were fixed for scanning electron microscope (SEM) observations using procedures identical to those used for TEM. For SEM imaging, the samples were dehydrated with ethanolic series, dried on critical point EM-CPD-030 (Leica, Heidelberg, Germany), and then sputter-coated with gold prior to examination (Schmidt et al., 2012*a*). The samples were examined under SEM JSM 6390 LV (JEOL Ltd. at 10 kV). Lead and copper were analyzed in the cell wall and cryofracture section for cytoplasm using SEM (NORAN System 7, Thermo Scientific, Rockford, IL, USA) coupled to an energy dispersive X-ray spectrometer (SEM-EDX), without post-fixing the samples in osmium tetroxide or gold coating.

Pigments Analysis

The content of photosynthetic pigments (chlorophyll *a* and phycobiliproteins) of *G. domingensis* was analyzed for the treatment group and control. Immediately after collection, the samples (fresh weight) were frozen by immersion in liquid nitrogen and kept at −40°C until ready for use. All pigments were extracted in quadruplicate as previously reported (Schmidt et al., 2010*b*).

Chlorophyll *a*

Chlorophyll *a* (Chl *a*) was extracted from approximately 1 g of tissue in 3 mL of dimethylsulfoxide (DMSO, Merck, Darmstadt, Germany) at 40°C, for 30 min, using a glass tissue homogenizer (Hiscox & Israelstam, 1979; Schmidt et al., 2010*c*). Pigments were quantified spectrophotometrically, according to Wellburn (1994).

Phycobiliproteins

About 1 g of algae material was ground to a powder with liquid nitrogen and extracted at 4°C in darkness in 0.1 M phosphate buffer, pH 6.4. The homogenates were centrifuged at 2,000 *g* for 20 min. Phycobiliprotein levels [allophycocyanin (APC), phycocyanin (PC), and phycoerythrin (PE)] were determined by ultraviolet-visible (UV-vis) spectrophotometry, and calculations were performed using the equations of Kursar et al. (1983).

Carotenoid Analyses

Carotenoids were extracted from samples (1.0 g fresh mass, $n = 4$) using hexane: acetone (1:1, v/v) containing 100 mg L^{-1} *tert*-butyl hydroxytoluene (BHT). Solutions were filtered through a cellulose membrane to remove particles, and the organosolvent extract was evaporated under N_2 flux. The residue was dissolved in hexane (3 mL). Prior to chromatographic analysis, 10% KOH in methanol (100 μL/mL) was added to 1 mL of the organosolvent extract to obtain complete carotenoid saponification, which allowed better identification of compounds by high performance liquid chromatography (HPLC). This solution was incubated (3 h in the dark at room temperature), followed by washing with distilled-deionized water (three times). The de-esterified extract was collected, concentrated under an N_2 flux, and resolubilized in hexane: acetone: BHT (100 μL) for further chromatographic analysis, as previously described (Kuhnen et al., 2009). A concentrated sample (10 μL, $n = 3$) was injected onto the liquid chromatograph (Shimadzu LC-10A, São Paulo, Brazil) equipped with a C_{18} reverse-phase column (Vydac 218TP54; 250 mm × 4.6 mm ∅, 5 μm, 30°C), protected by a 5 μm C_{18} reverse-phase guard column (Vydac 218GK54) and a UV-vis detector (450 nm). Elution was performed with MeOH: CH_3CN (90: 10, v/v) at a flow

Table 1. Growth Rates (GRs) of *G. domingensis* after 7 days of Exposure to Lead and Copper Treatments ($n = 4$, ± SD)*

	Control	5 ppm of Pb	10 ppm of Pb	5 ppm of Cu	10 ppm of Cu
GRs % day^{-1}	3.4 ± 1.5^{a}	2.4 ± 2.0^{b}	2.3 ± 1.0^{b}	-4.0 ± 1.5^{c}	-4.8 ± 1.0^{c}

*Superscript letters indicate significant differences according to Tukey's test ($p \leq 0.05$).

rate of 1 mL·min^{-1}. Carotenoid identification (β-carotene and lutein) was performed using retention times and co-chromatography of standard compounds (Sigma-Aldrich, St. Louis, MO, USA), as well as by analogy with other reports of carotenoid analysis by RP-HPLC-UV-vis under similar conditions (Scott & Eldridge, 2005; Hulshof et al., 2007). Carotenoid quantification was based on standard curves, employing the lutein standard curve (0.5–45 μg mL^{-1}; $y = 7{,}044x$; $r^2 = 0.999$) for lutein xanthophyll quantification and the β-carotene standard curve (0.01–12 μg mL^{-1}; $y = 1{,}019x$; $r^2 = 0.998$). The results were computed as means of triplicates and expressed as mean ± standard deviation (SD).

Biochemical Analyses

Glutathione reductase, NADH dehydrogenase activities, and protein content were assessed in the samples. To accomplish this, samples from the control and treated *G. domingensis* groups were homogenized in 20 mM phosphate buffer, pH 7.4, and centrifuged at 1,000 × *g* for 10 min at 4 °C. The supernatant was separated and used for assessing glutathione reductase activity and protein content.

Glutathione Reductase Assay

Glutathione reductase (GR) activity was determined by the method described by Carlberg and Mannervik (1985). The rate of GSSG reduction was indirectly determined through monitoring the NADPH disappearance at 340 nm. Results are expressed as μg/mg protein.

Sample Preparations for Measuring the NADH Dehydrogenase Activity

Samples from the control and treated *G. domingensis* groups were homogenized in 10 volumes of 50 mM phosphate buffer, pH 7.4, containing 0.3 M sucrose, 5 mM 3-morpholinopropanesulfonic acid (MOPS), 1 mM EGTA, and 0.1% bovine serum albumin (BSA). The homogenates were centrifuged at 1,000 × *g* for 10 min at 4°C; the pellet was then discarded, and the supernatants were used for measuring NADH dehydrogenase activity (Schmidt et al., 2012*c*).

Determination of NADH Dehydrogenase Activity

NADH dehydrogenase activity was assessed in supernatants by the rate of NADH-dependent ferricyanide reduction at 420 nm (1 mM^{-1} cm^{-1}), as previously described in Cassiana and Radi (1996). This method was slightly modified, as detailed in a previous report by Latini et al. (2005). Enzyme activity was calculated as nmol/minute/mg protein.

Protein Determination

The amount of protein in the samples was determined according to Lowry et al. (1951), using BSA as the standard.

Data Analysis

Data were analyzed by unifactorial ANOVA and Tukey's *a posteriori* test using the Statistica software (Release 6.0), considering $p \leq 0.05$. Statistical comparisons were performed to evaluate the variations in growth rates, concentration of photosynthetic pigments, and biochemical response under lead and copper treatments.

Results

Growth Rates

After 7 days of cultivation, control, lead- and copper-treated *G. domingensis* samples were compared, and a significant difference in GRs ($p < 0.02$) was observed. Control plants showed the highest GRs of 3.4% day^{-1} and increasing formation of lateral pigmented branches at the end of the experiment (Fig. 1a). Plants exposed to lead or copper treated at 5 or 10 ppm showed lower GRs than control (Table 1), and biomass loss was observed in copper-treated plants. Relative reduction of lateral branches and depigmentation were observed in lead-treated plants (Figs. 1b, 1c) and copper-treated algae (Figs. 1d, 1e).

Observations under LM and Cytochemistry

Control cells of *G. domingensis* stained with TB-O showed metachromatic reaction in the cell walls, indicating the presence of acidic polysaccharides, such as agar (Fig. 2a). Similar to control, transversal section of lead- (Figs. 2b, 2c) or copper-treated (Figs. 2d, 2e) plants at 5 and 10 ppm also revealed metachromatic reaction. Changes in mucilage and cell wall thickness are summarized in Table 2.

Control samples stained with PAS exhibited a strong reaction in the cell wall and mucilage layer, indicating the presence of cellulosic compounds and neutral polysaccharides (Fig. 2f). Positive reaction for neutral polysaccharides was detected in the cytoplasm with a notable presence of floridean starch grains, the particular reserve substance of red algae (Fig. 2f). Cortical cells exhibited an abundance of starch grains, which decreased in number toward the subcortical cells (Fig. 2f). Lead-treated cortical cells seemed to present amounts of starch grains similar to those of control plants, but subcortical cells showed higher density (Figs. 2g, 2h). The increasingly thick mucilaginous layer in plants exposed to lead can be seen in Figures 2g and 2h. On the other hand, plants treated with copper (5 ppm) showed a

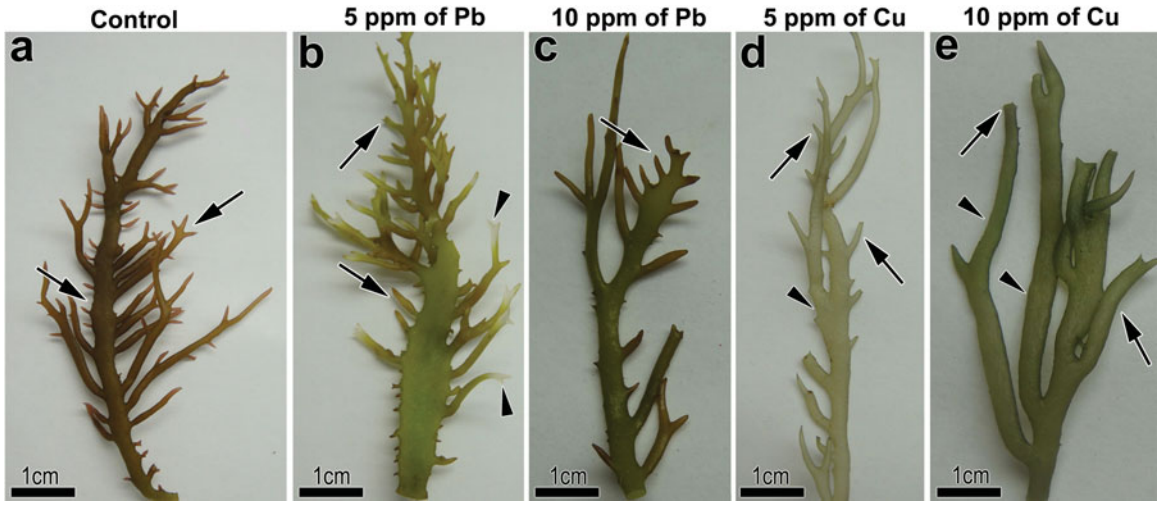


Figure 1. Morphological structure of *G. domingensis* after 7 days of exposure to lead and copper. (**a**) Control treatment showing natural red color and large quantity of dichotomies (arrows). (**b**, **c**) Lead treatment at 5 and 10 ppm, respectively, showing reduction of dichotomies (arrows) and thallus bleaching (arrowheads). (**d**, **e**) Copper treatment at 5 and 10 ppm, respectively, showing extreme thallus bleaching (arrowheads) and delicate tip portions (arrows).

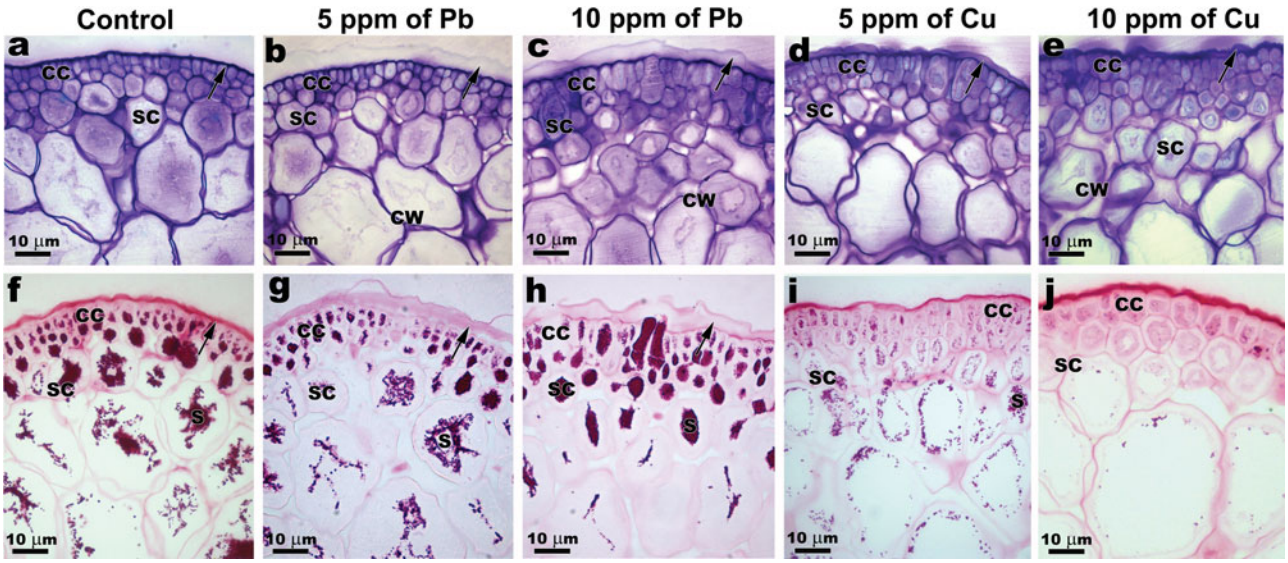


Figure 2. Light microscopy of thallus transversal sections of *G. domingensis* after 7 days of exposure to lead and copper treatments. Sections stained with TB-O. (**a**) Metachromatic reaction of control cell wall of cortical cell (CC) and subcortical cell (SC). (**b**, **c**) Lead-treated and (**d**, **e**) copper-treated plants with metachromatic reaction in the cell walls (CW) of cortical and subcortical cells. Sections stained with PAS. (**f**) Positive PAS reaction evidencing floridean starch grains (S) in the cells, cell wall and mucilage (arrow) in control plants. (**g**, **h**) Lead-treated plants with large quantity of starch grains in cortical and subcortical cells. Note the increasing mucilage thickness (arrow). (**i**) 5 ppm copper-treated plant with decreasing floridean starch grains in the cortical and subcortical cells. (**j**) 10 ppm copper-treated plant with dramatic reduction of floridean starch grains in cortical and subcortical cells.

reduction of floridean starch grains (Fig. 2i). Interestingly, plants treated with copper at 10 ppm (Fig. 2j) showed few floridean starch grains in either cortical or subcortical cells, as well as mucilage reduction (Fig. 2j).

Confocal Laser Scanning Microscopy

Confocal microscopy of *G. domingensis* under control treatment showed cortical cells with a large quantity of chloroplasts and high autofluorescence intensity (Fig. 3a). Both metal treatments led to reduction of autofluorescence (Figs. 3b–3e) compared to the control, but a more dramatic effect was detected under copper treatment (Figs. 3d, 3e) than lead (Fig. 3b, 3c). Chloroplast morphology was altered by heavy metal treatments, suggesting a susceptible toxic effect.

Scanning Electron Microscope

Under SEM, the surface topography of control *G. domingensis* showed regular undulations of cortical cell walls, indicat-

Table 2. Thickness Measurements (μm) of Mucilage and Cell Wall of *G. domingensis* Following TB-O Staining of Cortical and Subcortical Cells after 7 days of Exposure to Different Concentrations of Lead and Copper ($n = 60$ cells, $\pm$ SD).*

	Control	5 ppm of Pb	10 ppm of Pb	5 ppm of Cu	10 ppm of Cu
Mucilage	3.5 ± 1.0^{d}	2.0 ± 1.0^{e}	12.0 ± 2.0^{a}	6.0 ± 1.0^{c}	8.0 ± 3.0^{b}
Cell wall (cortical cell)	12.0 ± 3.0^{c}	14.0 ± 3.0^{b}	19.0 ± 5.0^{a}	14.0 ± 3.0^{b}	13.5 ± 1.8^{b}
Cell wall (subcortical cell)	3.5 ± 1.0^{c}	5.0 ± 1.0^{b}	14.0 ± 2.0^{a}	6.0 ± 2.5^{b}	5.5 ± 2.0^{b}

*Superscript letters indicate significant differences according to Tukey's test ($p \leq 0.05$).

ing a rough surface aspect (Fig. 4a). In contrast, plants treated with lead (Figs. 4b, 4c) and copper (Figs. 4d, 4e) showed a gradual decrease in rough surface, apparently as a result of heavy metal absorption. This result indicates that plants exposed to lead and copper probably undergo changes in the mucilage that coats the thallus.

The results of X-ray micronanalysis of *G. domingensis* can be considered qualitative. X-ray microanalysis of control cell wall surface (Table 3) revealed the presence of peaks of elements, e.g., calcium, carbon, iron, magnesium, oxygen phosphorus, and sulfur, but lead and copper were not detected on the surface of the control cell wall (Table 3). Lead and copper were found in the cell wall surface of *G. domingensis* treated with 5 and 10 ppm.

X-ray microanalysis of control cytoplasm did not reveal the presence of lead. However, copper was detected in the

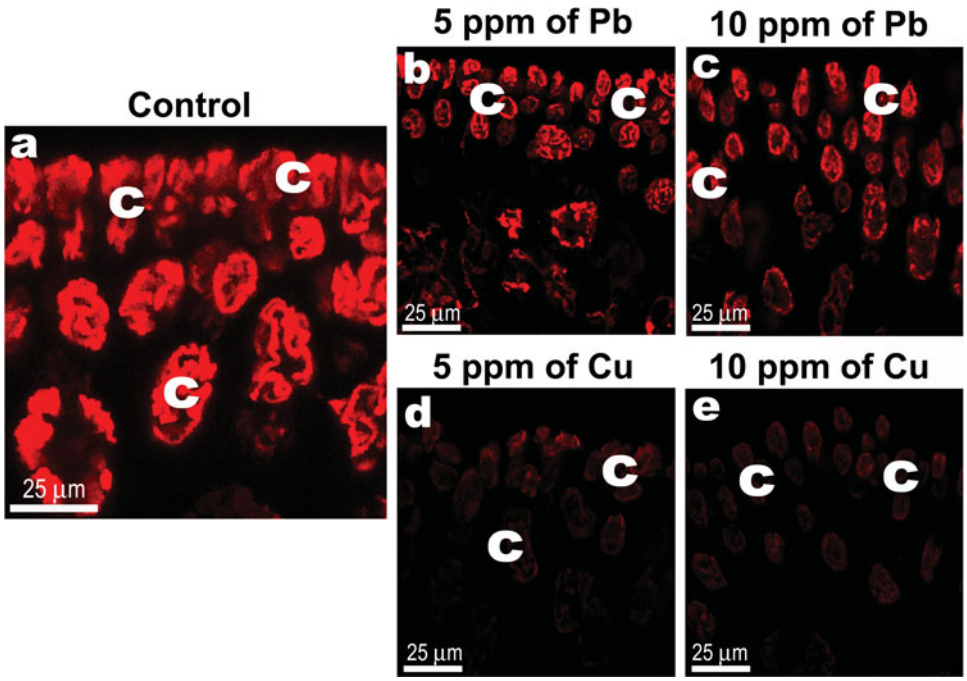


Figure 3. Confocal microscopy of thallus transversal sections of *G. domingensis* after 7 days of exposure to lead and copper treatments. (**a**) Control samples showing large amount of chloroplasts (C) and high autofluorescence. (**b**, **c**) Lead-treated plants showing chloroplast morphology and moderate autofluorescence. (**d**, **e**) Copper-treated plants showing low autofluorescence.

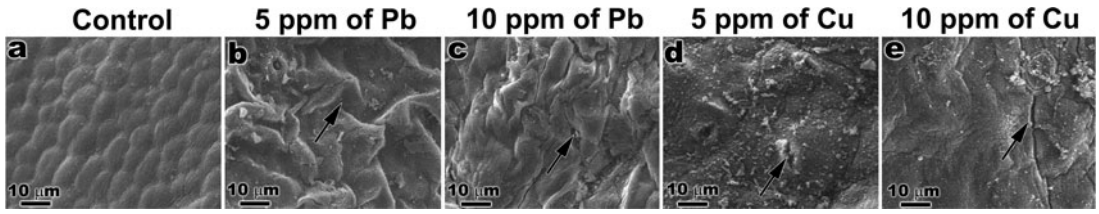


Figure 4. Scanning electron microscopy images of *G. domingensis* after 7 days of exposure to lead and copper treatments. (**a**) Detail of control surface topography of cortical cell walls showing a shriveled aspect. (**b–e**) Topography of lead- and copper-treated plants showing a rough and irregular surface (arrows) after heavy metal treatment.

Table 3. X-Ray Microanalysis of Cell Wall of *G. domingensis* after 7 Days of Exposure to Lead and Copper Treatments.*

	Control	5 ppm of Pb	10 ppm of Pb	5 ppm of Cu	10 ppm of Cu
C	57.7	55.0	63.0	60.0	54.0
O	31.3	30.12	28.09	20.7	31.0
Mg	0.56	0.33	0.27	0.23	0.64
P	1.37	0.97	0.94	1.28	3.14
S	6.18	3.48	3.87	2.95	3.92
Ca	1.30	3.55	0.97	4.57	4.74
Fe	1.65	0.69	0.55	9.83	3.0
Pb	—	5.88	2.49	—	—
Cu	—	—	—	2.96	3.00

*C, carbon; O, oxygen, Mg, magnesium; P, phosphorus; S, sulfur; Ca, calcium; Fe, iron; Pb, lead; Cu, copper ($n = 4$).

Table 4. X-Ray Microanalysis of Cytoplasm of *G. domingensis* after 7 Days of Exposure to Lead and Copper Treatment.*

	Control	5 ppm of Pb	10 ppm of Pb	5 ppm of Cu	10 ppm of Cu
C	60.79	62.68	59.79	61.39	56.26
O	32.07	32.69	32.12	20.77	28.72
P	0.1	0.23	—	0.47	0.73
S	6.93	4.07	7.81	9.99	4.6
Ca	0.11	0.33	0.28	4.61	4.94
Fe	—	—	—	0.55	1.74
Pb	—	—	—	—	—
Cu	—	—	—	2.17	3.01

*C, carbon; O, oxygen, Mg, magnesium; P, phosphorus; S, sulfur; Ca, calcium; Fe, iron; Pb, lead; Cu, copper ($n = 4$).

cytoplasm of *G. domingensis* for both concentrations used in the study (see Table 4).

Transmission Electron Microscope

When observed by transmission electron microscopy, *G. domingensis* cells of the cortical region were somewhat vacuolated and filled with numerous organelles, especially chloroplasts (Fig. 5a), surrounded by a thick cell wall (Fig. 5b), and the chloroplasts assumed the typical internal unstacked organization of red algae, i.e., evenly spaced thylakoids (Fig. 5c). Electron-dense lipid droplets described as plastoglobuli were observed between the thylakoids (Fig. 5c).

After lead treatment, *G. domingensis* showed irregularly shaped cortical cells and increasing cell wall thickness with concentric layers of microfibrils (Fig. 5d). Inside the cell wall, holes were observed (Fig. 5d). A large amount of floridean starch grains was also verified in the cortical cells (Figs. 5d–5h). The presence of electron-dense precipitates could be seen near the chloroplast and starch grains, probably lead (Figs. 5f–5h). Chloroplasts showed obvious changes in ultrastructural organization with irregular morphology (Figs. 5i–5l). Thylakoids were disrupted and swollen (Figs. 5i–5l), as well as increasing plastoglobuli with different shapes (Figs. 5i–5k). Some chloroplasts became vacuolated (Fig. 5k), and a large quantity of concentric membranes was observed in the cytoplasm (Fig. 5l).

Copper treatments caused more dramatic changes than lead, with cortical cells showing large reduction in cytoplasmic cell volume (Figs. 6a–6h). Some cells showed a large vacuole (Fig. 6b). The cell wall had more compact aspect (Figs. 6a, 6b, 6e), and the organelles were completely destroyed (Figs. 6c, 6d). Some floridean starch grains were observed in the cytoplasm (Figs. 6e–6h). The chloroplasts were degenerated and disrupted (Figs. 6g, 6h).

Pigments

Lead and copper treatments significantly decreased the content of photosynthetic pigments in *G. domingensis* (Table 5). Lead and copper treatments decreased Chl *a* levels compared to control algae, as well as the amounts of phycobiliprotein contents (APC, PC, and PE) and total carotenoids. The values of concentration of all photosynthetic pigments were significantly different between control and heavy metal-treated plants.

The amount of carotenoids found in *G. domingensis* plants exposed to copper and lead was determined by HPLC (Table 5). Chromatographic analysis allowed the identification of lutein and *trans*-β-carotene in both treatments. The plants treated with lead showed an increase in lutein content in contrast to that detected for copper-treated plants. Moreover, similar amounts of *trans*-β-carotene were found for control and plants exposed to lead. Copper dramatically reduced *trans*-β-carotene content.

Table 5. Changes of Photosynthetic Pigments [Means ± SD μg/g (Fresh Weight)] of *G. domingensis* after 7 Days of Exposure to Lead and Copper Treatments ($n = 4$).*

Treatments	Chl *a*	APC	PC	PE	Lut	trans-β-car
Control	605 ± 1.5^{a}	550 ± 4.2^{a}	350 ± 1.5^{a}	300 ± 1.2^{a}	0.75 ± 0.05^{b}	0.49 ± 0.03^{a}
5 ppm of Pb	405 ± 2.5^{c}	51 ± 1.0^{b}	30 ± 5.0^{b}	90 ± 2.0^{b}	0.91 ± 0.02^{a}	0.49 ± 0.09^{a}
10 ppm of Pb	470 ± 4.0^{b}	26 ± 2.0^{d}	15 ± 1.0^{c}	80 ± 4.0^{c}	0.97 ± 0.06^{a}	0.53 ± 0.08^{a}
5 ppm of Cu	70 ± 5.5^{e}	40 ± 4.0^{c}	13 ± 1.5^{c}	50 ± 1.5^{d}	0.25 ± 0.02^{c}	0.02 ± 0.01^{b}
10 ppm of Cu	100 ± 2.5^{d}	35 ± 9.0^{c}	10 ± 0.05^{c}	40 ± 1.0^{e}	0.28 ± 0.08^{c}	0.00 ± 0.00

*Chl *a*, chlorophyll *a*; APC, allophycocyanin; PC, phycocyanin; PE, phycoerythrin; Lut, lutein; *trans*-β-car, *trans*-β-carotene. Superscript letters indicate significant differences according to Tukey's test ($p \leq 0.05$).

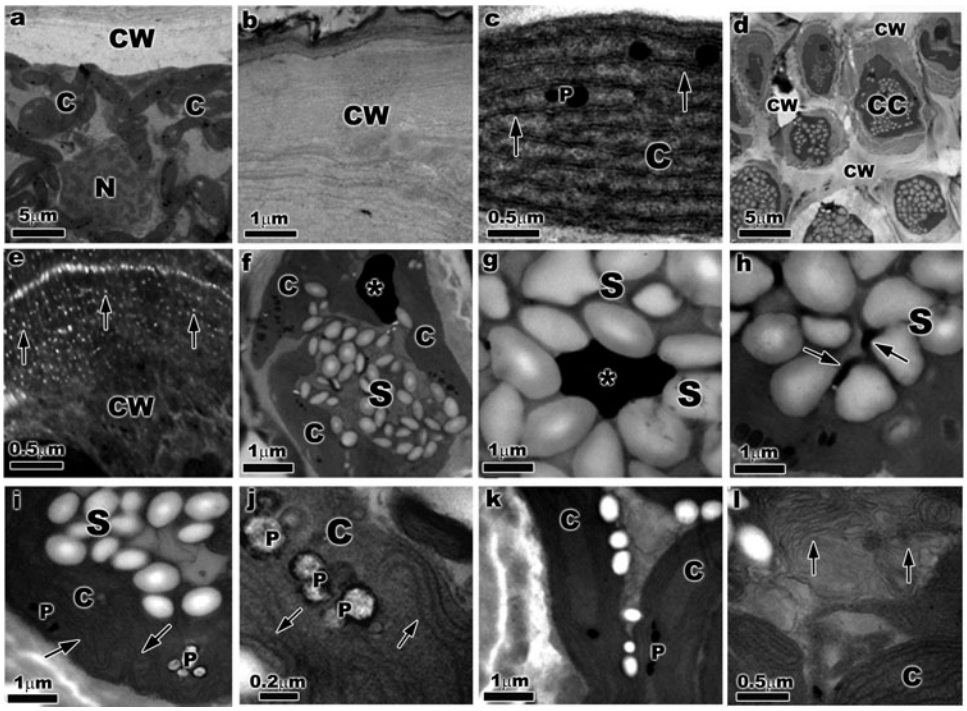


Figure 5. Transmission electron microscopy micrographic images of (**a–c**) *G. domingensis* control plants and (**d–l**) plants treated with 5 and 10 ppm of lead. (**a**) Detail of cortical cell with various chloroplasts (C) embedded in thick cell wall (CW) and the nucleus (N). (**b**) Magnification of the thick cell wall embedded in a mucilaginous substance. (**c**) Detail of chloroplast (C) of a cortical cell with intact thylakoids (arrows) and plastoglobuli (P). (**d**) Detail of cortical cells (CC) with a large quantity of floridean starch grains. (**e**) Observe the thickness in the cell wall with increase of concentric microfibril layers. (**f**) Note electron-dense precipitates (asterisk) between the chloroplasts. (**g**, **h**) Observe these precipitates (asterisk and arrows) near the starch grains. (**i**, **j**) Note the disrupted chloroplast, with some intact thylakoids (arrows) and the presence of plastoglobuli. (**k**) Detail of vacuolated chloroplast. (**l**) Note the presence of concentric membranes (arrows) near the chloroplasts. (**d**, **f**, **g**, **h**, **i**) Plants treated with 5 ppm of lead; (**e**, **j**, **k**, **l**) plants treated with 10 ppm of lead.

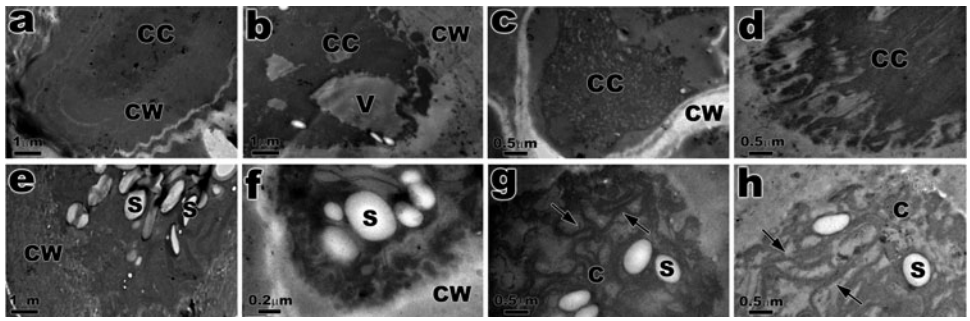


Figure 6. Transmission electron microscopy micrographic images of *G. domingensis* cultivated with 5 and 10 ppm of copper. (**a–d**) Detail of cortical cells showing the thickening of cell wall. Note the irregular outline and the presence of vacuoles (V). (**e**, **f**) Note the presence of some starch grains in cortical cells. (**g**, **h**) Detail of disrupted, vacuolated, and swollen chloroplasts (arrows). (**a**, **c**, **d**, **e**, **g**) Plants treated with 5 ppm of copper; (**b**, **f**, **h**) plants treated with 10 ppm of copper.

Biochemical Analyses

Lead-treated *G. domingensis* plants did not alter their protein content compared to control treatment (Fig. 7a). On the other hand, plants treated with copper showed a significant decrease ($P < 0.001$) (Fig. 7a). NADH dehydrogenase activity was lower in lead and copper treatments (Fig. 7b) with a major effect under copper treatment. The GR activity decreased after copper exposure (Fig. 7c) when compared with control and lead-treated plants (Fig. 7c).

Discussion

The present study shows changes in cellular architecture, ultrastructure, and cytochemistry, as well as metabolic response, of the agarophyte *G. domingensis* exposed individu-

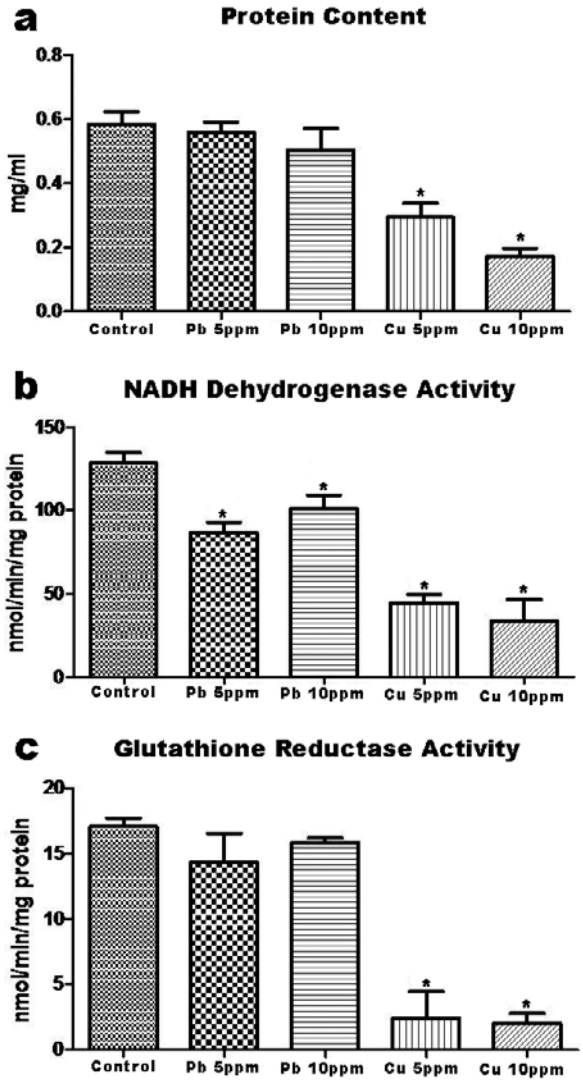


Figure 7. Biochemical responses of *G. domingensis* after 7 days of exposure to lead and copper treatments. (**a**) Protein content of *G. domingensis*. (**b**) Relative changes of mitochondrial NADH dehydrogenase activity of *G. domingensis*. (**c**) Glutathione reductase assay activity of *G. domingensis*. Means ± SD ($n = 4$). The asterisk indicates the statistical differences.

ally to 5 and 10 ppm of lead or copper. *G. domingensis* treated with copper showed a decrease in GRs, together with a reduction in algal biomass and, hence, negative growth rates, indicating that copper stress is a key factor limiting growth. However, while plants treated with lead showed a slower rate of growth than control, no reduction in algal biomass was evident. Ologuin et al. (2009) observed different results in *Gracilaria coronopifolia* and *Gracilaria eucheumoides* exposed to lead. Nonsignificant differences in GRs were observed in *G. coronopifolia*, while GRs in *G. eucheumoides* were inhibited. These results clearly indicate that the toxicity of lead is species dependent.

On the other hand, the bleaching and partial necrosis of the apical segments of *G. domingensis* during 7 days of cultivation under copper and lead could have resulted from changes in metabolism and DNA damage by the formation of pyrimidine dimers (Bouzon et al., 2012; Santos et al., 2012). Tissue deformation and necrosis were also observed in the carragenophyte *H. musciformis* after exposure to cadmium during 7 days (Bouzon et al., 2012) and *G. domingensis* exposed to UV radiation-B during 21 days (Schmidt et al., 2010*a*).

According to Visviki and Rachlin (1992), copper is an essential element for the photosynthetic electron transport system and a component or cofactor of several enzymes. The algae may actively exclude, or sequester, the copper to minimize toxic impact. The decrease in GRs observed in the *G. domingensis* samples studied may be related to the use of energy as an activation mechanism for the adaptation or repair of damage induced by high copper stress.

The cell wall of the control and heavy metal-treated plants of *G. domingensis* reacted positively to TB-O. This occurred because the cell wall contains sulfated polysaccharides in the molecular structure of agar. The metachromatic granulations were observed inside the cortical and subcortical cells of plants treated with lead and copper. Especially in vacuoles, these granulations appeared as reserves of complex polysaccharides and could be sites of accumulation that was subsequently incorporated into the cell wall and the mucilage. Overall, this finding indicates an increase in cell wall thickness, which was observed in cells after treatment with TB-O of *G. domingensis* exposed to lead and copper.

When analyzed under a TEM, the cell wall of *G. domingensis* control showed a microfibrillar texture with microfibrils structured in concentric layers with different degrees of compression. Observations of the cell wall of plants exposed to lead and copper showed an increase in that structural component. The increase in the thickness of the cell wall of *G. domingensis* exposed to lead and copper can be interpreted as a defense mechanism. The presence of precipitates in cytoplasm of *G. domingensis* exposed to lead (see Figs. 5f, 5g) could be associated with a metal-immobilizing function to minimize lead stress.

The analysis of the cell walls of treated plants by energy dispersive spectrometer made it possible to detect the presence of lead and copper, indicating the metal accumulation and metal binding in these structures. An increase in the presence of Ca and Fe was verified on the cell wall surface of plants treated with Cu. On the other hand, Ca was reduced in plants treated with 10 ppm of Pb. This change in cell wall surface composition with increasing concentration of Ca and Fe may be associated with a barrier against the presence of copper. The sulfated polysaccharides that compose the cell wall, such as agar and carrageenan, are responsible for most of the metal accumulation and metal binding (Bouzon et al., 2012). In the case of agar, metal binding is associated with the degree of sulfatation (Diannelidis & Delivopoulos, 1997; Bouzon et al., 2012; Santos et al., 2012). According the Sharma and Dubey (2005), mucilage binding restricts metal uptake and establishes an important barrier protecting the cells. The retention of Pb was observed in the cell wall and mucilage upon the binding of Pb to ion-exchangeable sites on the cell wall and extracellular precipitation, mainly in the

form of Pb carbonate deposited in the cell wall. The presence of carboxyl and sulfated groups in the cell walls of *G. domingensis* restricts the transportation of Pb via apoplast. On the other hand, Yruela (2005) describes many copper transporters already characterized in plant cells, such as those in the cell membrane (COPT1), cytosol (copper chaperones) and chloroplast (COPT3, PAA1, PAA2, and Cp CCS), mitochondria (COX17), and Golgi body (RAN1, chaperones and COPT5).

Cellular metabolism of *G. domingensis* exposed to lead and copper was modified. As a consequence, cell wall thickening was produced in both treatments, and this, in turn, resulted in copper-induced reduction in the synthesis of floridean starch grains. The decrease in floridean starch grains observed through TEM and LM in plants exposed to copper may be related to a change in the route of biosynthesis of starch enzymes of the Calvin cycle, possibly by activating the degradation pathways. However, Schmidt et al. (2010*a*) observed one increase in floridean starch grains in *G. domingensis* after exposure to UV radiation-B, and Bouzon et al. (2012) observed a decrease in starch grains in *H. musciformis* after treatment with different concentrations of cadmium.

In red algae, the thylakoids that are not associated with each other are free in the chloroplasts. The chloroplasts of *G. domingensis* control showed a structure typical of red algae, having one peripheral thylakoid surrounded by parallel thylakoids. The number of parallel thylakoids is variable, and this number mainly depends on the spatial location of the cell in the algae. In contrast, the chloroplasts of *G. domingensis* exposed to lead showed structural changes, including disrupted and swollen thylakoids.

However, the plants treated with copper showed more dramatic chloroplast alteration. Similar results were observed with the red macroalgae *Ceramium ciliatum* exposed to copper, where the chloroplast appeared with disrupted thylakoids and an increase in plastoglobuli volume (Diannelidis & Delivopoulos, 1997) and the green algae *Enteromorpha flexuosa* (Andrade et al., 2004). Finally, when analyzed by TEM, *G. domingensis* exposed to lead revealed an increase in the number of plastoglobuli in the chloroplast. This increase in the number of lipids can be considered as a change in metabolism, as observed in *G. domingensis* treated with lead.

In the present study, Chl *a* contents of *G. domingensis* showed more resistance upon exposure to lead in comparison to copper, which showed a reduction in concentration by 85%. These results can be associated with changes in chloroplast organization, i.e., a drastic reduction in autofluorescence, as shown by confocal microscopy. Other investigations of red macroalgae have shown an increase in Chl *a* contents after copper exposure, including, for example, *G. tenuistipitata* (Pinto et al., 2011). However, in *Gracilaria lemaneiformis*, Xia et al. (2004) observed that Chl *a* content was not altered after exposure to copper.

The amounts of phycobiliproteins decreased in *G. domingensis* treated with lead and copper. The phycobiliproteins are located in the phycobilisomes outside the chloroplast thylakoids. In red algae, PE is used during the acclimation process; therefore, it is located more externally in the phycobilisomes (Talarico, 1996). Our results demonstrated that phycobiliprotein levels, including APC, PC, and PE, decreased in *G. domingensis* after lead and copper treatments. These molecules absorb solar energy, transferring it to the reaction center of photosystem II, where Chl *a* is excited by the flow of electrons (Gantt, 1981). According to Xia et al. (2004), a high concentration of copper altered phycobilisome structure, and these changes resulted in a decline of absorbed light energy, thus inhibiting photosynthesis. We found a decrease in phycobiliprotein levels, similar to the findings of Xia et al. (2004), who studied the red macroalgae *G. lemaneiformis* cultivated with copper during 4 days. This indicates that lead and copper strongly inhibited the accumulation of phycobiliproteins.

Carotenoid contents in *G. domingensis* were also altered by lead and copper treatment. The contents of carotenoids, mainly lutein, remarkably increased in lead-treated *G. domingensis* plants, suggesting that this metal induces the accumulation of those pigments, along with the decrease in Chl *a* and phycobiliprotein contents. This result may be interpreted as a protective acclimation mechanism to prevent the toxic effects of lead. Collén et al. (2003) and Pinto et al. (2011) observed similar results in *G. tenuistipitata* exposed to copper and cadmium, with the increase of lutein and β-carotene. In contrast, copper-treated *G. domingensis* plants decreased their carotenoid contents by 78%, and *trans*-β-carotene was not detected in copper-treated samples.

According to Collén et al. (2003), heavy metal in exposed algae induces the production of reactive oxygen species (ROS). The ROS induce changes in several molecules, such as lipids, proteins, and nucleic acid. In *G. tenuistipitata* treated with cadmium and copper, one increase in the amount of saturated and monounsaturated fatty acids (Pinto et al., 2011). However, in the same study, a reduction of the levels of polyunsaturated fatty acids was observed. As a strategy to prevent the effects of ROS, photosynthetic organisms produce antioxidant defenses, such as flavonoids, tocopherols, carotenoids, and enzymes. Collén et al. (2003) observed that treatment of *G. tenuistipitata* with copper led to an increase in the enzymatic activity of catalase, ascorbate peroxidase, and superoxide dismutase, while treatment with cadmium only increased catalase activity.

The significant reduction in protein content observed in copper-treated *G. domingensis*, when compared with control and lead-treated plants, indicates that copper inhibits protein metabolism. Similar results were observed by Collén et al. (2003) in *G. tenuistipitata* treated with cadmium and copper. They proposed that the decreased content of protein could have been caused by toxic effects to protein or amino acid synthesis, as well as nitrate reduction. In our study, treatment with both copper and lead blocked mitochondrial NADH dehydrogenase activity, including that of complex I. Under these conditions, complex I of the respiratory chain has been reported as one of the major sites of ROS generation in mitochondria under physiological conditions (Boveris et al., 1976; Chen et al., 2003), and an overt

ROS production is observed when this redox center is blocked (Turrens, 1997; Turrens & Boveris, 1980). Furthermore, ROS formed at the NADH dehydrogenase site of complex I are released into the mitochondrial matrix (Chen et al., 2003), thereby eliciting oxidative damage to mitochondrial enzymes, including the complexes of the respiratory chain, enzymes of the Krebs cycle, and several other sensitive proteins, as well as mtDNA (Bandy & Davison, 1990; Zhang et al., 1990; Hausladen & Fridovich, 1994).

However, our study found that cellular antioxidant potential could be characterized by a significant increase in nonenzymatic antioxidant defenses, such as the increase in lutein content in the presence of high lead concentration, while antioxidant enzymatic activities were essentially unchanged by copper treatment.

In summary, the present study demonstrates that lead and copper negatively affect various morphological, physiological, and biochemical processes in *G. domingensis*. This became obvious after exposure of plant samples to concentrations of 5 and 10 ppm under laboratory conditions over a 7-day experimental period, resulting in reduction in growth rates, ultrastructural damage observed primarily in the internal organization of chloroplasts, increased cell wall thickness and plastoglobuli. Based on the relative toxicity profile of the two heavy metals studied in *G. domingensis*, copper is more toxic than lead.

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